

Orange City Batteries — Nagpur

Battery Marketplace Website Full Technical & SEO Specification Document

Prepared for	OCB Nagpur — Development & SEO Team
Market scope	Nagpur, Maharashtra (hyperlocal only)
Benchmarked against	BatteryBoss.in, BatteryBhai.com
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1. Executive Summary

Orange City Batteries (OCB) is converting from a lead-generation website into a transactional, hyperlocal battery marketplace built to out-rank and out-convert every player currently visible for battery searches in Nagpur — specifically **BatteryBoss.in** and **BatteryBhai.com**, both national aggregators with no physical presence in the city.

The central strategic fact driving this entire specification: **neither competitor has a real shop, technician, or stock in Nagpur**. They operate a dropship/dealer-routing model — an order placed online gets forwarded to a local partner for fulfillment. OCB already is that local partner in substance (real shop in Pardi, 4.9★ Google rating, 58 reviews, 6 AM emergency service) but currently lacks their online conversion mechanics: visible pricing, exchange-offer discounting, cart/checkout, and deep SEO catalog coverage.

This document specifies a website that combines **their proven online conversion mechanics** with **OCB's unmatched local trust and speed**, engineered from the ground up for maximum organic search ranking in Nagpur — not national reach.

2. Competitor Analysis

2.1 BatteryBoss.in

- National multi-brand aggregator — car, bike, inverter, truck, industrial, lithium, gel batteries (Exide, Amaron, SF Sonic, Luminous, Livfast, Bosch, AC Delco)
- Two-tier pricing on every SKU**: "With Old Battery" (exchange) price vs "Without Old Battery" price — exchange price typically 30–40% cheaper. This is their single biggest conversion lever.
- MRP shown with large "% OFF" badges; all prices GST-inclusive, no hidden costs
- Free shipping + free installation via local dealer network, same-day to 24-hour fulfillment
- COD + card/UPI/EMI
- Nagpur appears only as a city-selector option — **no physical footprint**
- SEO: brand × vehicle-model matrix pages (e.g. `/car-batteries/kia`), limited city-level landing pages

2.2 BatteryBhai.com

- "India's No. 1 online multi-brand battery store" — car, inverter, two-wheeler, generator, solar, UPS, stabilizers
- Same exchange-value + free delivery/installation conversion model
- Claims 100+ cities via a **"Partner With Us" dealer network** — local shops register and "pay only when you sell." This means local Nagpur competitors may already be getting recruited as BatteryBhai's Nagpur fulfillment arm — a live competitive risk.
- Engagement tools: Battery Health Check, Power/Load Calculator — useful for lead capture even pre-purchase
- Claims 99% installation within 24 hours of order
- SEO: category pages per vehicle type (4-wheeler, truck, 2-wheeler, solar)

2.3 Pattern Common to Both — and What It Means for OCB

Pattern	Implication for OCB
No real local presence in Nagpur	Local trust + speed is a structural advantage they cannot copy without opening a real shop
Exchange-price psychology drives conversion	Must show exchange vs non-exchange price prominently on every product, not behind a quote request
Vehicle-model-first navigation	Battery Finder tool must lead directly into priced SKUs, not just a recommendation
Recruiting local dealers as fulfillment partners	Time-sensitive — move first, or a Nagpur rival becomes "the BatteryBhai/BatteryBoss dealer" and captures both online and offline demand
Deep brand × vehicle-model SEO catalog	OCB replicates this pattern but adds a locality dimension neither competitor has: brand × vehicle-model × Nagpur locality

3. Strategic Positioning for Nagpur Domination

Lever	BatteryBoss / BatteryBhai	OCB Nagpur
Local presence	None — dealer-routed fulfillment	Real shop, real stock, real technicians
Pricing transparency	Shown online with exchange discount	Match exactly — MRP, exchange price, non-exchange price, %OFF badge on every product
Speed	Few hours to 24 hrs	Beat it — 6 AM emergency service + real-time locality-based delivery ETA
Trust signals	Generic "100,000+ customers	4.9★ Google + JustDial reviews, locality-specific and stacked visibly on every page

	across India"	
SEO depth	Brand × vehicle-model, limited city pages	Brand × vehicle-model × Nagpur locality — a depth of coverage they cannot replicate without local knowledge

Design principle governing every decision in this document: every page, feature, and content choice is filtered through — "does this help win the Nagpur customer, today?" This is not a national marketplace build.

4. Website Objectives & Success Criteria

- Rank in the top 3 organic results for "car battery Nagpur," "battery price Nagpur," and 15+ locality-specific terms within 90 days of launch
- Convert visitors directly into paid orders via cart/checkout — not solely WhatsApp/call leads
- Load fast enough to pass Core Web Vitals on mobile (majority of local searches are mobile) — see Section 11
- Every page must be crawlable and indexable with content and price present in the initial HTML response (no client-side-only rendering of price-critical content)
- Match or exceed BatteryBoss/BatteryBhai's price-transparency and exchange-offer UX
- Surface OCB's local trust signals (reviews, real address, real technicians) more prominently than any competitor page can

5. Technology Stack Specification

Layer	Choice	Justification (ranking-first)
Frontend framework	Next.js 14 (App Router, TypeScript)	Native SSR/SSG/ISR — every product and locality page renders as complete HTML at build/request time. Price and content sit in the initial DOM, eliminating the single biggest local-SEO risk: search engines indexing a page before client-side JS has fetched the price.
Database	PostgreSQL	Relational structure matches the domain exactly: brand → product → vehicle compatibility → locality → order.
Database hosting	Supabase	Postgres + Auth + Storage bundled — one service instead of three, faster for a 3-person team to manage.
ORM	Prisma	Type-safe schema and migrations as the catalog scales from tens to hundreds of SKUs.
Styling	Tailwind CSS	Minimal CSS payload — page weight is a direct Core Web Vitals / ranking factor.
Payments	Razorpay, called server-side only (Next.js Route Handler)	Best India SMB support for COD + UPI + card + EMI; keys never exposed to the browser.
Hosting	Vercel (or Netlify with the Next.js runtime adapter)	Native ISR support, automatic image optimization, edge caching — all directly improve page speed and thus ranking.
Sitemap & structured data	<code>next-sitemap</code> + JSON-LD (Product, LocalBusiness, AggregateRating, BreadcrumbList schemas)	Enables rich results — star ratings and price snippets directly in the Google search listing, a visible differentiator over both competitors' plain listings.
Image delivery	Next.js Image component + Supabase Storage or a CDN	Automatic responsive sizing/lazy-loading — directly impacts Largest Contentful Paint (a Core Web Vital).
Analytics	Google Analytics 4 + Google Search Console (mandatory), optionally Microsoft Clarity for heatmaps	Search Console is required to monitor indexing status and ranking — non-negotiable for an SEO-first build.

6. System Architecture



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Database webhook → rebuild hook
(price/stock change triggers
incremental page regeneration)

Price-freshness mechanism: when a dealer updates a price or stock level in Supabase, a database webhook fires a Vercel/Netlify deploy hook, triggering an incremental rebuild of the affected pages within roughly 30–60 seconds — pages remain static (fast, SEO-safe) but are never stale.

7. Database Schema Specification

Table	Purpose	Key Fields
brands	Battery brand master	id, name, slug, logo_url
products	Battery SKUs	id, brand_id, model_name, slug, category (car/bike/inverter/truck/industrial/UPS), battery_type (flat/tubular/VRLA/lithium/gel), capacity_ah, warranty_months, mrp, price_with_exchange, price_without_exchange, installation_included, stock_status, images[], description
vehicle_compatibility	Vehicle make/model fitment per product	id, product_id, vehicle_make, vehicle_model, year_from, year_to
localities	Nagpur locality master — the core SEO asset	id, name, slug, pincode, delivery_eta_hours
product_locality_availability	Stock & ETA per product per locality	id, product_id, locality_id, in_stock, eta_override
orders	Customer orders	id, customer_name, phone, email, address, locality_id, payment_method, payment_status, razorpay_order_id, status, total_amount, exchange_opted, created_at
order_items	Line items per order	id, order_id, product_id, quantity, unit_price
reviews	Aggregated trust signals	id, product_id (nullable for site-wide), rating, text, author_name, source (google/justdial/site), created_at

Minimum target locality list at launch (expandable by SEO team): Pardi, Sadar, Dharampeth, Sitabuldi, Hingna, Wardha Road, Manish Nagar, Trimurti Nagar, Civil Lines, Ramdaspath — plus full coverage as keyword research identifies additional high-intent localities.

8. Site Map & URL Structure

URL Pattern	Purpose	Rendering
/	Homepage — value proposition, trust signals, featured products	Static / ISR
/products/[slug]	Individual product page with full pricing, JSON-LD Product schema	Static / ISR
/category/[car bike inverter truck]	Category listing pages	Static / ISR
/brand/[brand-slug]	All products for a given brand	Static / ISR
/locality/[locality-slug]	Hyperlocal SEO page — "Battery Price in Pardi, Nagpur" — one per locality	Static / ISR
/brand/[brand]/[vehicle-model]	Brand x vehicle-model matrix pages (mirrors competitor depth, Nagpur-scoped)	Static / ISR
/battery-finder	Interactive vehicle-based battery finder, links directly into priced SKUs	Client-interactive island
/cart , /checkout	Cart and checkout flow	Client-interactive
/blog/[slug]	SEO content — buying guides, seasonal content	Static / ISR
/about , /reviews , /contact	Trust and conversion support pages	Static
/sitemap.xml , /robots.txt	Auto-generated, always current	Build-time generation

9. Functional Specification (Page by Page)

9.1 Homepage

- Hero section: clear value proposition ("Nagpur's fastest battery delivery, real shop, real technicians")
- Trust bar: Google rating (4.9★/58), JustDial verified badge, "real Nagpur shop since [year]" messaging
- Category quick-links: Car / Bike / Inverter / Truck / Industrial
- Featured/best-selling products with visible pricing
- Battery Finder CTA
- Locality coverage map or list ("We deliver to: Pardi, Sadar, Dharampeth...")
- **JSON-LD** LocalBusiness schema present

9.2 Product Page

- MRP, price with exchange, price without exchange, %OFF badge — all visible without interaction
- Warranty, capacity (Ah), battery type
- Compatible vehicle list (if applicable)
- Delivery ETA, dynamically reflecting the customer's selected/detected locality
- Add to Cart, with exchange toggle
- **JSON-LD** Product + AggregateRating schema
- Unique, keyword-optimized title tag and meta description generated per product

9.3 Locality Page (highest strategic priority)

- H1: "Battery Price in [Locality], Nagpur"
- Locality-specific delivery ETA messaging
- Full product listing filtered to that locality's availability
- Locally-flavored content: nearby landmarks, typical delivery time, any locality-specific notes
- **JSON-LD** LocalBusiness schema with areaServed set to the locality

9.4 Battery Finder

- Vehicle make/model/year selector
- Returns compatible products with pricing — links directly to product pages, not just a recommendation summary

9.5 Cart & Checkout

- Cart: item list, quantity, running total
- Checkout form: name, phone, address, locality selector (validated against the locality table)
- Payment method: COD or Razorpay (UPI/card/EMI)

- Order confirmation: on-screen + WhatsApp message + email
- Exchange pickup scheduling captured at checkout if exchange opted

9.6 Blog / Content Hub

- Seasonal content ("Why batteries fail in Nagpur summer heat")
- Buying guides per vehicle category
- Evergreen page: "Battery Price in Nagpur 2026" — updated regularly, internally linked from every locality page

10. SEO Specification

10.1 Technical SEO — Non-Negotiable Requirements

- All pricing and product content must be present in the server-rendered HTML — verified via "View Source," not just DevTools after JS execution
- Auto-generated `sitemap.xml` including every product, locality, brand, and blog page, resubmitted to Search Console on every deploy
- `robots.txt` correctly configured — no accidental blocking of `/products/` or `/locality/` paths
- Canonical tags on every page to prevent duplicate-content issues across category/brand/locality overlaps
- Mobile-first responsive design, tested on real devices, not just browser emulation
- HTTPS enforced site-wide; no mixed-content warnings
- Structured data validated with Google's Rich Results Test for every page template before launch

10.2 On-Page SEO

- Unique title tag + meta description per product, locality, brand, and category page (no templated duplicates)
- H1 present once per page, matching primary target keyword
- Internal linking: every locality page links to relevant products; every product page links to relevant localities where it's available
- Image alt text descriptive and keyword-relevant (not "IMG_0234")
- Breadcrumb navigation with BreadcrumbList schema

10.3 Local SEO

- Google Business Profile: full category setup, complete service-area list, Q&A seeding, regular photo uploads, active review-request funnel
- NAP (Name, Address, Phone) consistency enforced across website, Google Business Profile, JustDial, and any other directory listing — inconsistency directly damages local ranking
- JustDial optimization: verified badge displayed on-site, catalog/pricing synced if the subscription tier supports it, reviews cross-surfaced on the website
- Locality × brand × vehicle-type keyword-to-page mapping maintained and expanded by the SEO lead on an ongoing basis

10.4 Content & Link Building

- Ongoing content calendar: seasonal guides, locality-targeted posts, comparison content
- Backlinks from Nagpur business directories, automotive forums, and local city blogs/news outlets
- Social proof presence: WhatsApp Business, Instagram, Facebook — local community engagement

11. Performance & Core Web Vitals Targets

Metric	Target	Why it matters
Largest Contentful Paint (LCP)	< 2.5s on mobile	Direct Google ranking factor; also the single biggest driver of bounce on slow mobile connections
Interaction to Next Paint (INP)	< 200ms	Determines perceived responsiveness of cart/checkout interactions
Cumulative Layout Shift (CLS)	< 0.1	Prevents mis-clicks on price/CTA elements — directly affects conversion
Time to First Byte (TTFB)	< 600ms	Achieved via static/ISR rendering + edge CDN caching
Lighthouse Performance Score	90+ on mobile	Overall page-speed health check, tracked per release

12. Payments & Checkout Specification

1. Customer selects product, chooses "with exchange" or "without exchange" price
2. Enters locality/pincode — sees live delivery ETA sourced from the locality table
3. Cart → Checkout: name, phone, address, delivery slot
4. Payment: COD (available immediately) or Razorpay UPI/card/EMI (requires merchant KYC completion before go-live)
5. Order confirmation: WhatsApp message + email, order persisted to the orders table
6. If exchange opted, old-battery pickup is scheduled as part of the same order flow

Dependency: Razorpay merchant KYC must be initiated early — approval can take several business days and blocks the online-payment (non-COD) path.

13. Security & Compliance

- All API keys (Razorpay, Supabase service role) stored server-side only, never exposed to client-side JavaScript
- Row-level security policies enabled on Supabase tables containing customer data
- HTTPS enforced site-wide with valid SSL (handled automatically by Vercel/Netlify)
- Input validation and sanitization on all checkout form fields
- Rate limiting on the checkout API route to prevent abuse
- GST-inclusive pricing displayed transparently, matching invoicing requirements

14. Content & Asset Deliverables Checklist

Deliverable	Owner
Full product catalog data (all SKUs, brands, prices, exchange prices, warranty terms)	OCB business team
Product photography (each SKU, minimum 2 angles)	OCB business team
Vehicle compatibility data per product	OCB business team / SEO research
Full Nagpur locality target list with delivery ETAs	SEO lead
Keyword-to-page mapping document	SEO lead
Initial blog content calendar (minimum 10 articles)	SEO lead
Logo, brand colors, favicon, OG share image	OCB business team
Google Business Profile access	OCB business team
JustDial subscription access/credentials	OCB business team
Razorpay merchant account (KYC completed)	OCB business team

15. Full Project Deliverables List

1. Production-deployed Next.js marketplace website on `ocbnagpur.in`
2. Complete Postgres database (Supabase) with schema per Section 7, populated with full catalog
3. Working cart → checkout → order flow, COD and Razorpay both functional
4. Minimum 20 locality pages live at launch, expandable via CMS/database entry (no code deploy needed to add a locality)
5. Full brand × vehicle-model matrix pages for all carried brands
6. Battery Finder tool linking directly into priced SKUs
7. Auto-generated, always-current sitemap.xml submitted to Google Search Console
8. JSON-LD structured data validated on every page template
9. Google Business Profile fully optimized and verified
10. JustDial listing NAP-aligned and cross-linked
11. Analytics: Google Analytics 4 + Search Console connected and reporting
12. Performance report confirming Core Web Vitals targets (Section 11) are met on production
13. Handover documentation: admin/catalog management guide, deployment guide, environment variable reference

16. Implementation Phases & Timeline

Phase	Duration	Scope
Phase 0 — Foundation	Week 1	Repo/infra setup, database schema live, locality & keyword research delivered, Razorpay KYC initiated
Phase 1 — MVP Marketplace	Weeks 2-4	Priced product pages, cart/checkout (COD), 15-20 locality pages, Battery Finder, Google Business Profile optimization
Phase 2 — Catalog Depth & Trust	Weeks 5-7	Full brand × vehicle-model matrix, exchange pickup scheduling, reviews aggregation, blog content, JustDial sync, Razorpay online payment live
Phase 3 — Scale & Retention	Week 8+	B2B/fleet portal, WhatsApp catalog sync, referral program, geo-fenced Google Ads, full pincode-level locality coverage

17. Team Roles & Responsibilities

Role	Responsibilities
Developer 1	Site architecture, database integration, catalog/admin workflows, performance tuning
Developer 2	Pricing/exchange logic, Razorpay integration, checkout flow, QA/testing
SEO Lead	Keyword mapping, locality page content, Google Business Profile + JustDial optimization, backlinks, content calendar
Shared (all)	Content QA, launch review, post-launch ranking and conversion monitoring

18. KPIs & Success Metrics

Metric	Target
Organic ranking	Top 3 for "car battery Nagpur," "battery price Nagpur," + 15 locality terms within 90 days
Indexing speed	100% of product/locality pages indexed within 14 days of publish
Conversion rate	Cart-to-order rate on new pricing pages, tracked against pre-launch WhatsApp-only baseline
Review growth	Combined Google + JustDial review count +50% within 90 days
Revenue mix	Online-originated orders as % of total monthly orders, tracked monthly from launch
Core Web Vitals	All targets in Section 11 met on 100% of key page templates

19. Risks & Mitigations

Risk	Mitigation

Price-critical content rendered client-side only, harming SEO	Enforced architecturally via SSR/ISR — verified in QA by checking raw HTML source, not just the rendered browser view
Competitors recruit a local Nagpur dealer as their fulfillment partner first	Prioritize Phase 1 launch speed + Google Business Profile/JustDial trust work in parallel — time-sensitive
Small team capacity across dev + SEO alongside other commitments	Phases are independently shippable; Phase 1 alone validates the core hypothesis
Razorpay KYC delays blocking online payment	Initiate onboarding in Phase 0; COD remains fully functional in the interim
Catalog data (prices, images, vehicle compatibility) incomplete at launch	Treat Section 14 checklist as a hard gate before Phase 1 go-live

20. Acceptance Criteria for "100% Working, Best-SEO" Sign-off

The website is considered complete and ready for launch only when every item below is verifiably true:

- ☐ Every product and locality page returns full content (including price) in the raw server-rendered HTML — verified via "View Source"
- ☐ Sitemap.xml is live, correct, and submitted to Google Search Console with zero indexing errors reported
- ☐ JSON-LD structured data on every page template passes Google's Rich Results Test with no errors
- ☐ Lighthouse mobile Performance score ≥ 90 on homepage, a sample product page, and a sample locality page
- ☐ Core Web Vitals targets (Section 11) met on production, confirmed via PageSpeed Insights / Search Console Core Web Vitals report
- ☐ Cart → Checkout → Order confirmation flow completes successfully end-to-end for both COD and Razorpay
- ☐ Minimum 20 Nagpur locality pages are live, unique, and internally linked from relevant product/category pages
- ☐ Google Business Profile and JustDial listing NAP data matches the website exactly
- ☐ Google Analytics 4 and Search Console are connected and reporting live data
- ☐ No mixed-content or SSL warnings anywhere on the site
- ☐ Admin/catalog handover documentation delivered and walked through with the team

— End of Specification —